

Using Singular Value Decomposition (SVD) to Solve Linear Regression

Singular Value Decomposition (SVD) is a powerful linear algebra tool that can be used to solve linear regression problems in a more numerically stable way. Here's how it works:

1. Problem Setup: The Linear Regression Model

In linear regression, the goal is to find a parameter vector θ that minimizes the squared difference between the predicted values $\mathbf{X}\theta$ and the observed values $\mathbf{y}$. Mathematically, this is represented as:

$$\theta = \arg \min_{\theta} \|\mathbf{X}\theta - \mathbf{y}\|^2$$

The standard approach to solving this problem is to use the **Normal Equation**:

$$\theta = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

However, directly computing the inverse of $\mathbf{X}^T \mathbf{X}$ can be numerically unstable, especially if $\mathbf{X}^T \mathbf{X}$ is nearly singular or poorly conditioned. This is where SVD comes in.

2. Perform SVD on $\mathbf{X}$

SVD decomposes the matrix $\mathbf{X}$ into three matrices:

$$\mathbf{X} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$$

Where:

- $\mathbf{U}$ is an $m \times m$ orthogonal matrix (columns are left singular vectors of $\mathbf{X}$).
- $\mathbf{\Sigma}$ is an $m \times n$ diagonal matrix with non-negative real numbers on the diagonal (the singular values of $\mathbf{X}$).
- $\mathbf{V}^T$ is the transpose of an $n \times n$ orthogonal matrix (rows are right singular vectors of $\mathbf{X}$).

3. Rewrite the Normal Equation Using SVD

The Normal Equation for linear regression is:

$$\theta = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Using the SVD of $\mathbf{X}$, we can substitute $\mathbf{X} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$ into the equation:

$$\theta = ((\mathbf{U} \mathbf{\Sigma} \mathbf{V}^T)^T (\mathbf{U} \mathbf{\Sigma} \mathbf{V}^T))^{-1} (\mathbf{U} \mathbf{\Sigma} \mathbf{V}^T)^T \mathbf{y}$$

Expanding this:

$$\theta = (\mathbf{V}\mathbf{\Sigma}^T\mathbf{U}^T\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T)^{-1}\mathbf{V}\mathbf{\Sigma}^T\mathbf{U}^T\mathbf{y}$$

Since $\mathbf{U}$ is an orthogonal matrix ($\mathbf{U}^T\mathbf{U} = \mathbf{I}$, where $\mathbf{I}$ is the identity matrix), this simplifies to:

$$\theta = (\mathbf{V}\mathbf{\Sigma}^T\mathbf{\Sigma}\mathbf{V}^T)^{-1}\mathbf{V}\mathbf{\Sigma}^T\mathbf{U}^T\mathbf{y}$$

Next, we use the fact that $\mathbf{V}$ is also orthogonal ($\mathbf{V}^T\mathbf{V} = \mathbf{I}$):

$$\theta = \mathbf{V}(\mathbf{\Sigma}^T\mathbf{\Sigma})^{-1}\mathbf{V}^T\mathbf{V}\mathbf{\Sigma}^T\mathbf{U}^T\mathbf{y}$$

Simplifying further:

$$\theta = \mathbf{V}(\mathbf{\Sigma}^T\mathbf{\Sigma})^{-1}\mathbf{\Sigma}^T\mathbf{U}^T\mathbf{y}$$

4. Simplification Using the Diagonal Structure of $\mathbf{\Sigma}$

$\mathbf{\Sigma}^T\mathbf{\Sigma}$ is a diagonal matrix containing the squares of the singular values of $\mathbf{X}$. The inverse of this diagonal matrix is straightforward to compute by taking the reciprocal of each non-zero singular value. Let $\mathbf{\Sigma}^{-1}$ represent the diagonal matrix where each diagonal element is the reciprocal of the corresponding singular value. Then:

$$(\mathbf{\Sigma}^T\mathbf{\Sigma})^{-1}\mathbf{\Sigma}^T = \mathbf{\Sigma}^{-1}$$

Thus, the expression for θ simplifies to:

$$\theta = \mathbf{V}\mathbf{\Sigma}^{-1}\mathbf{U}^T\mathbf{y}$$

5. Interpretation and Advantages

This final expression for θ avoids directly computing the inverse of $\mathbf{X}^T\mathbf{X}$, which can be numerically unstable. Instead, it leverages the structure of the SVD to provide a more stable and efficient solution:

- **Orthogonality:** The orthogonal matrices $\mathbf{U}$ and $\mathbf{V}$ simplify the computation.
- **Diagonal Matrix Inversion:** Inverting the diagonal matrix $\mathbf{\Sigma}^T\mathbf{\Sigma}$ is straightforward and avoids the potential pitfalls of inverting large, dense matrices.
- **Numerical Stability:** SVD is particularly useful when $\mathbf{X}$ is ill-conditioned (i.e., when the columns of $\mathbf{X}$ are nearly linearly dependent).

This method is widely used in practical applications, especially when dealing with large datasets or when the matrix $\mathbf{X}^T\mathbf{X}$ is nearly singular, providing a robust way to compute the regression coefficients θ .